RYAN **HUANG**

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SUMMARY

I'm a systems architect experienced in scaling platforms (1→100) within big tech and leading 0→1 development in nimble startups. Expert at the intersection of distributed systems and AI infrastructure, I have a proven track record of engineering high-performance, cost-efficient ML systems that bridge the gap between research and production. Leveraging deep expertise across applications, infrastructure, and LLM, I drive the technical roadmap for next-generation distributed training and inference ecosystems.

EXPERIENCE

2022 - Present, Staff ML Engineer, AI Infra - Noetik, Seattle, WA

Tech Lead specializing in multimodal LLM infrastructure at a biotech AI research startup. Built the company's ML platform from the ground up, supporting AI research and production applications across developer workspaces, distributed training, large-scale batch inference, real-time serving, and agentic workflows.

- Built a PyTorch distributed training framework supporting FSDP, context parallelism, and tensor parallelism.
- Build large-scale LLM inference systems, and pipeline for LLM fine-tune (DPO/LORA) and quantization.
- Accelerated multimodal data loading through distributed file caching and an RDMA transfer engine.
- Optimized LLM serving with SGLang, quantization, speculative decoding, and prefill–decode disaggregation.
- Optimized custom CUDA/CUTLASS kernels to accelerate both training and inference workloads.

2020 - 2022, Sr. Software Engineer, Machine Learning Platform – Uber, Seattle, WA

- Build unified ML platforms for Uber (Built on kubernetes CRDs and operators).
- Designed and implemented a control plane to streamline ML model lifecycles (training, evaluation, monitoring).
- Enhanced ML developer experience through custom CLI tools for infrastructure interaction.
- Built a distributed orchestration framework to improve reliability of training pipelines on Spark and Ray.

2017 - 2020, Sr. Software Engineer, Amazon Transcribe – AWS, Seattle, WA

- Led the development of Amazon Transcribe (ASR) from scratch, creating multi-tenant cloud services
- Build real-time streaming service speech recognition, bidirectional streaming using HTTP/2 and GRPC.
- Led a team to develop applications for special domains, e.g., medical and call center analytics.
- Built full-stack ML platform using AWS Lambda and ECS.

2017 - 2017, Software Engineer – Thomson Reuters, Sunnyvale, CA

- Built an A/B testing platform leveraging Spark, Kafka, Cassandra, and Redis for data processing and analytics.

2014-2017, CPU Physical Design Engineer – Broadcom, Santa Clara, CA

- Physical design of ARM v16 CPU core, timing closure, place & route, and CAD tools, etc.

EDUCATION

2014 - Master of Science (Electrical & Computer Engineering), University of Florida - FL, United States

2013 - Bachelor of Science (Micro-electronics), UESTC - Chengdu, China

PATENT

No. US10777186B1: Streaming Real-time Automatic Speech Recognition Service.